

Age Prediction from fMRI Connectivity Data

—Using Autoencoder and Regularized Regression

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Abstract

The objective of this project is to predict each individual's age based on the fMRI data provided by Healthy Brain Network (HBN) and a metadata that includes demographic features. The fMRI data focused on each participant's brain activity during the resting-state. To have a better understanding of the large number of the features, Principal Component Analysis (PCA) and autoencoder were used for dimension reduction. We applied regularization regression models and further explored the machine learning approach—XGBoost to enable performance comparison. To enhance model performance, we utilized neural networks with techniques such as early stopping, grid search, the Adam optimizer, and dropout regularization. The model performances were evaluated by comparing the metrics of Rooted Mean Squared Error (RMSE) and Coefficient of Determination (R^2).

Introduction

As people grow older, changes occur in the way different brain regions communicate with each other. The fMRI (resting-state functional MRI) data measures each individual's brain activity when they are relaxed. Studies have shown that brain connectivity patterns change over time. As a result, there has been growing interest in using the brain scan data to predict age. This could help to monitor brain health and to detect early brain issues if the predictions are accurate and reliable.

This project is focusing on brain connectivity features which are recorded in the fMRI data to predict age. Different brain activity patterns reflect how the different brain regions interact with each other, and therefore may provide insights into how brain activity changes over time as well as identifying the connection features that are mostly related to aging.

Data Cleaning

The dataset includes two main components: functional MRI (fMRI) data and metadata (demographic information). A total of 1,104 participants are available. Each participant has a 200×200 functional connectivity matrix stored in an individual .tsv file, along with metadata such as sex, ethnicity, and race.

To process the fMRI data, we extracted the upper triangle of each connectivity matrix (excluding the diagonal), resulting in 19,900 correlation coefficients per participant. These values were flattened and combined into a single data frame of size $1,104 \times 19,901$, where each row corresponds to one participant. The participant ID was extracted from each .tsv filename and used as the row identifier.

To deal with missing values, we handled numeric and categorical variables separately. For example, in the case of missing body mass index (BMI) values, since BMI distributions differ significantly between males and females, we imputed missing BMI values using the mean BMI stratified by sex. For categorical variables, missing values (NAs) were treated as a separate category labeled “Missing”. All numeric metadata variables (excluding the 19,900 fMRI correlation features) were standardized to have zero mean and unit variance.

After preprocessing, the fMRI feature matrix and the cleaned metadata were merged into a single dataset for subsequent modeling. To ensure compatibility with machine learning and deep learning models, all categorical variables were one-hot encoded. This was implemented using the `dummyVars()` function in R, which automatically drops the first level of each factor to avoid multicollinearity. The same transformation process was applied to the test dataset. Importantly, to prevent data leakage, imputations and scaling were performed using statistics (e.g., means) calculated from the training data only.

EDA

This section provides an overall summary of the features from the metadata’s training set. Exploratory visualizations are used to show the distributions as well as relationships associated with age as guides for feature selection for modeling.

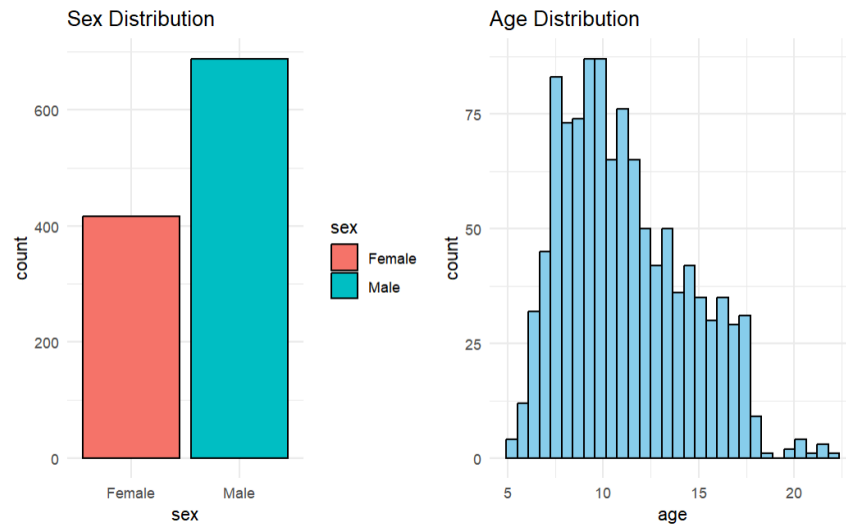


Figure 1: Sex and Age Distribution

The bar graph at the left shows the sex distribution. This dataset includes more males than females where there are 416 females and 688 males. The right histogram on the right shows the distribution of age. The peak indicates that the majority of the participants are approximately around the age of 9 to 11 years. There are a small number of participants younger than 7 or older than 15. Overall, the distribution of age is right skewed and the majority of the participants are teenagers.

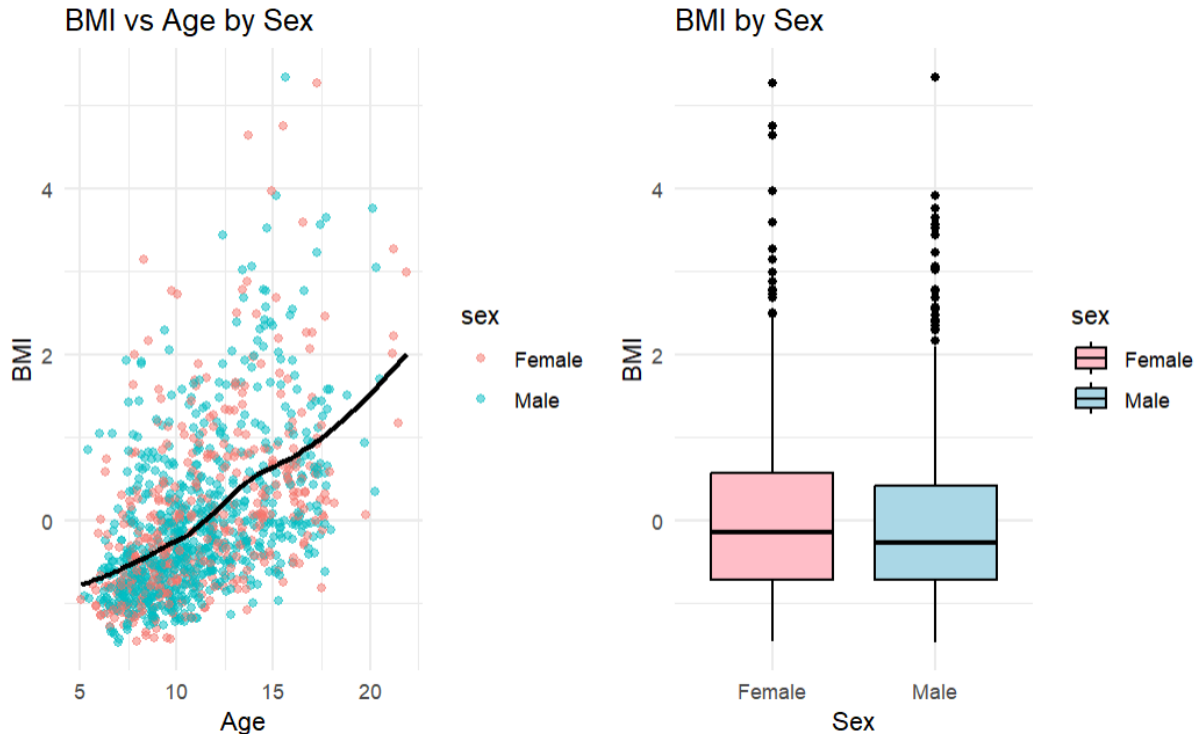


Figure 2: BMI visualizations

The scatter plot at the left shows the relationship between BMI versus Age where pink points indicate female, blue points for male. It has an overall upward trend for both males and females and it is shown by the black trend line. In other words, both males and females shared similar patterns in the relationship between BMI and Age.

The right box plot shows the overall distribution of BMI by sex. There is no big difference in the median BMI values of both female and male groups, but with females slightly higher. The spread of BMI in both groups are also similar and there are few outliers in the two groups as well but again, there seems to be no large difference in BMI values between males and females.

Methodology

We first split original data into a training set and a testing set based on the 8:2 proportion. The features from the metadata were standardized (zero mean, unit variance) in order to eliminate scale differences between variables. To address the high dimensionality of the fMRI features (~19,900), we tried both Principal Component Analysis (PCA) and autoencoder for dimensionality reduction to the training data. Then, based on the compressed datasets (PCA

version and autoencoder version), we fitted ridge, lasso, and elastic net regressions. By comparison, we can see that applying autoencoder to the training data set produced better results compared to those with PCA (Table 1 and Table 2).

Given that the autoencoder demonstrated superior performance compared to PCA (likely due to its ability to capture complex nonlinear structures in high-dimensional fMRI data), we further explored models that combined autoencoder-based dimensionality reduction with Lasso, Ridge, and Elastic Net regression. These regularized linear models were chosen for their strengths in handling high-dimensional, potentially collinear predictors: Lasso performs variable selection by enforcing sparsity, Ridge stabilizes coefficient estimates in the presence of multicollinearity, and Elastic Net offers a balance between the two.

Results

1. Regularized Regression on PCA

After applying PCA to the high-dimensional fMRI data, we retained 559 principal components to preserve 95% explained variance. Although dimensionality was substantially reduced compared to the original feature space, the resulting number of components is still relatively large. Therefore, it is reasonable that we selected regularized regression methods such as Ridge, Lasso, and Elastic Net to further address potential overfitting and multicollinearity among the transformed features. Root Mean Square Error and R-Squared are used to measure and compare the model performance. Ideally we expect lower RMSE and higher R^2 . Table 1 shows the results of five prediction models.

Table 1: Results for each model under PCA

Model (PCA)	RMSE	R^2
Ridge Regression	2.214	0.552
Lasso Regression	2.250	0.537
Elastic Net	2.276	0.526

As Table 1 shows, the ridge regression model has Test RMSE=2.214 and $R^2=0.552$, performing the best prediction performance among three models.

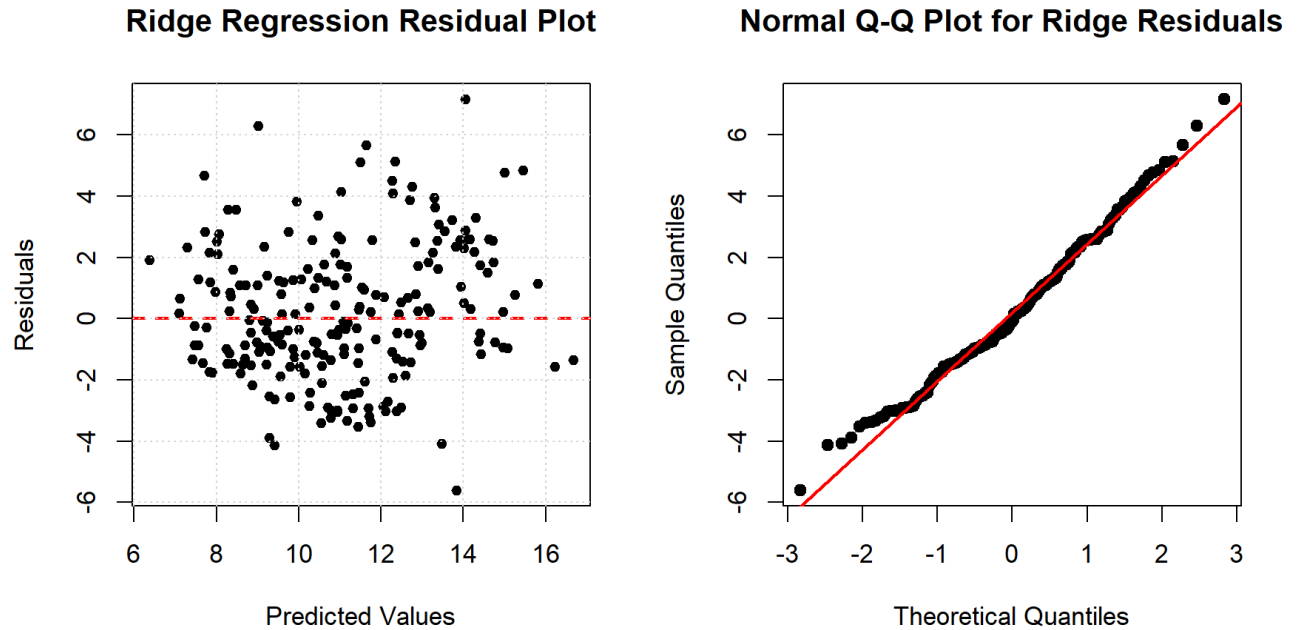


Figure 3: Residual plot and QQ plot of Ridge (PCA)

Figure 3 illustrates that the residuals are evenly distributed around zero across the range of predicted ages, which indicates that the model captures the underlying patterns in the data without apparent systematic bias. The QQ plot demonstrates conformity to the theoretical distribution, with residuals closely following the diagonal reference line throughout the entire range and minor deviations at the tails. This confirms that the normality assumption is satisfied. In summary, these diagnostics provide evidence that the ridge regression model is well-specified for this dataset, with error terms that are independently and identically distributed as required for valid statistical inference.

2. Regularized Regression on Autoencoder

According to Plis et al. (2014)^[5], autoencoders outperform Principal Component Analysis (PCA) in neuroimaging tasks. Therefore, we adopted autoencoder-based dimensionality reduction in our study. To optimize the autoencoder architecture, we conducted a grid search over latent

dimensions, hidden units, and dropout rates using early stopping and the Adam optimizer, minimizing the mean squared reconstruction loss. The best-performing configuration was found to be a latent dimension of 256, hidden units of 256, and a dropout rate of 0. Test set performance metrics for the resulting models are summarized in Table 2.

Table 2 Results for Autoencoder with each model

Model (Autoencoder)	RMSE	R^2
Ridge	1.824	0.673
Lasso	1.848	0.665
Elastic Net	1.777	0.690

Test RMSE is 1.777 which means that the average prediction error is approximately 1.777 years, and it is relatively low compared to the result of Ridge with PCA. Test $R^2 = 0.690$ means the model explains about 67% of the variance in age, suggesting a strong predictive ability. Elastic Net combines both L1 (Lasso) and L2 (Ridge) penalties, which allows it to perform variable selection while maintaining stability in the presence of multicollinearity. Given the high-dimensionality of the fMRI data, Elastic Net is well-suited because it can both shrink uninformative coefficients and retain correlated predictors rather than arbitrarily excluding them.

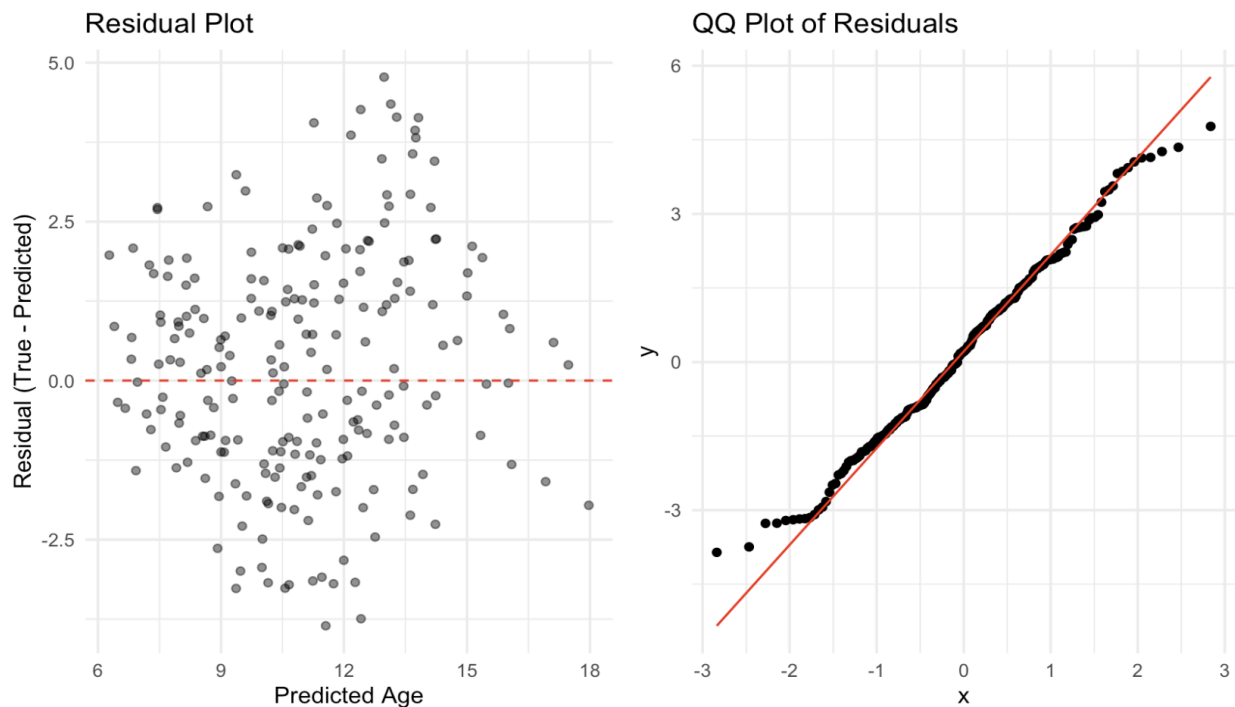


Figure 4: Residual plot and QQ plot of Elastic Net (Autoencoder)

The residual plot of the Elastic Net model shows that residuals are evenly scattered around zero across the range of predicted ages, with no apparent pattern or systematic deviation. This

indicates that the model captures the underlying structure of the data well, with no major violations of linearity or homoscedasticity. Overall, the residual distribution supports the adequacy and reliability of the Elastic Net model. The QQ plot shows that residuals from the Elastic Net model are approximately normally distributed, with minor deviations at the tails. This suggests that the normality assumption holds reasonably well for model evaluation.

Conclusion

After evaluating several machine learning approaches such as autoencoder plus XGBoost, and PCA plus regularized regressions etc., the best results on the validation set were produced by the combination of autoencoder and Elastic Net regression, where it has the lowest RMSE (1.777) and highest R^2 (0.690). The autoencoder first compressed the fMRI data into lower dimensional space (256) which helped to reduce the risk of overfitting as well as redundancy. The results from the net elastic regression suggest that brain connectivity carries significant signals related to an individual's age.

Reference

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Appendix

1. EDA of age with study site and sex

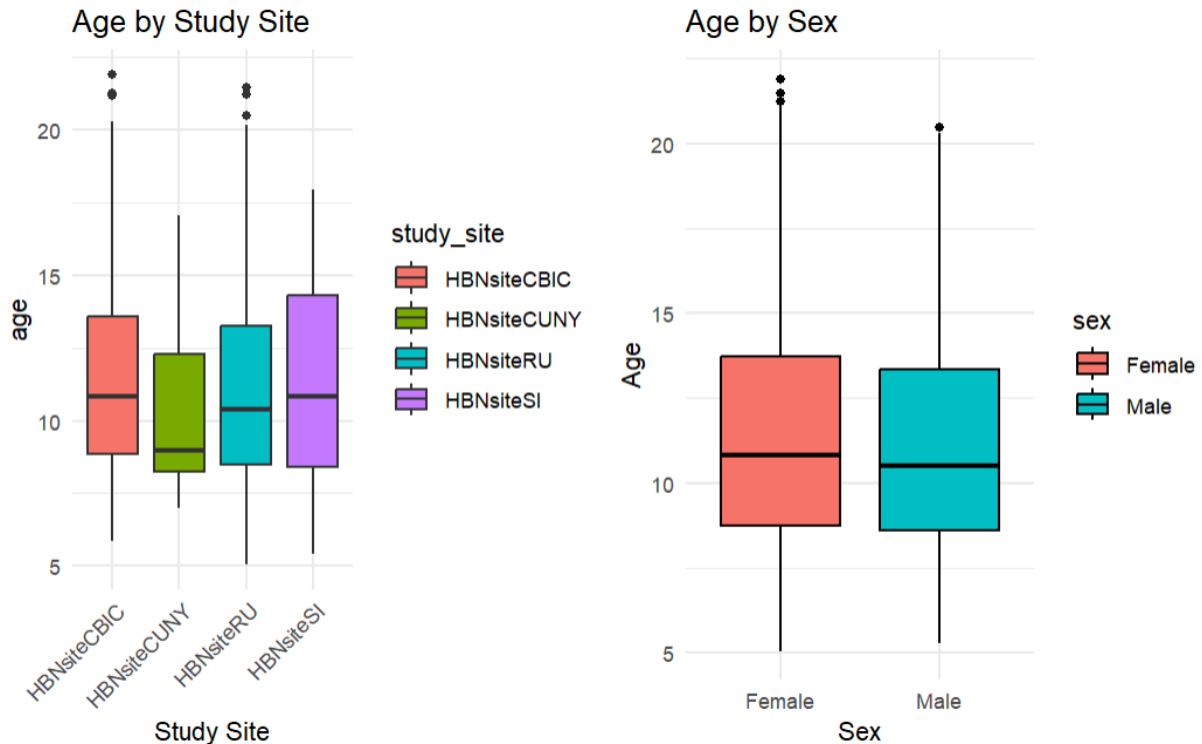


Figure 5: Distribution of Age by Study Site and Sex

The left box plot compares the age across different study sites. City College of New York has the lowest median of age. Other three study sites (Staten Island, Rutgers University, and CitiGroup Cornell Brain Imaging Center) have similar median age and they have more older participants included. In addition, the age ranges of the different study sites also varied, which indicates that study sites might be a covariate that influences age distribution. The right box plot shows the comparison of age across females and males. The median ages overall have a small difference where females have slightly higher median age. The age ranges for both males and females are wide where there are few females with age above 20.

2. Background information EDA

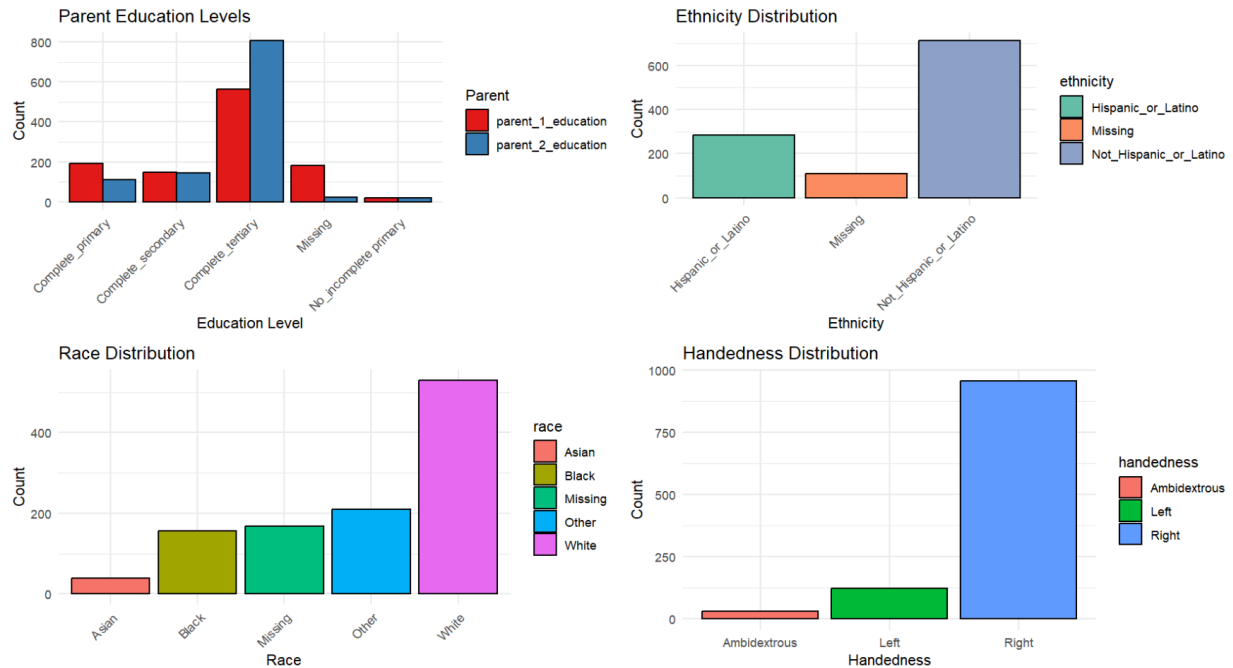


Figure 6: Distributions of Background Information

The above figure shows the overall summary of family and background variables. The top left plot shows the distribution of each participant's parent education levels. Majority of the participants have both parents completed tertiary (college) education, where parent 2 seemed to have a higher education level compared to parent 1.

The top right plot shows the ethnicity distribution. The distribution indicates that most of the participants are not Hispanic or Latino, while smaller percentages reported as Hispanic or Latino and few missing data exist.

The bottom left plot shows the race distribution of the data. Majority of the racial group reported is white, followed by Other, Missing, Black, and Asian. The diversified race identities may provide some insights into this study, however there are quite few missing values in race, which may require further exploration.

The bottom right plot shows the handedness distribution. Most of the participants are right handed. Only a small portion of the participants reported themselves as left handed or ambidextrous. Since there is a strong skew in the portion of the right handed population, this variable might have limitations in predicting age.

3. EDA with mental health data

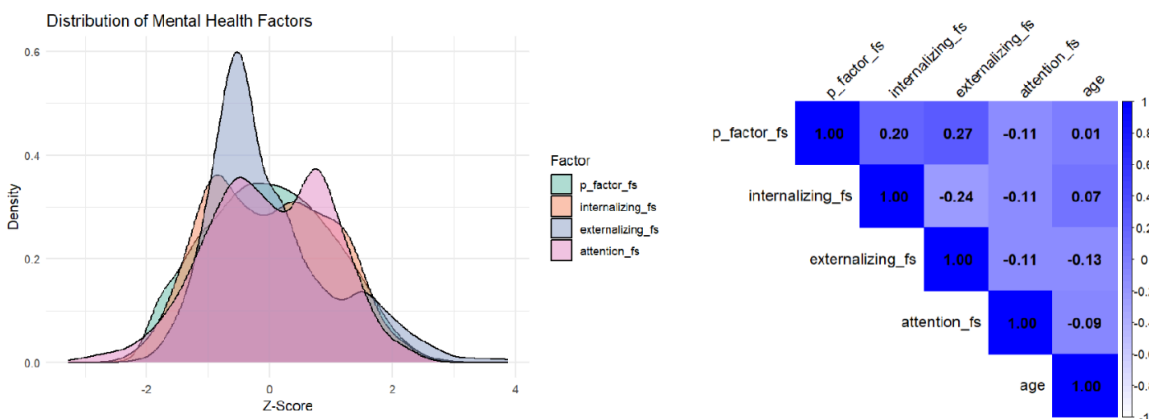


Figure 7: Distributions and Correlations of Mental Health Factors

The above density plot shows the summary of the distributions of the mental health factors. The four mental health factors are general factors (common thread to mental health disorders), internalizing factors (reflect emotional problems), externalizing factors (behavioural issues), and attention factors (inattention and hyperactivity). Notice that the externalizing factor has the narrowest distribution with the highest peak, which suggests lower variability. The remaining factors all have moderate spread, and the attention factor seemed to have a slightly right skewed distribution.

The above correlation matrix illustrates pairwise correlations between the mental health factors as well as age. The mental health factor that has the strongest correlation with age is the externalizing factor and they are negatively correlated (-0.13). However, this factor is not strongly correlated with age. Hence, from the above result, it can be concluded that the four mental health factors have weak correlations with age.

4. XGBoost on Autoencoder

We also want to Test the performance of machine learning on an autoencoder, so we choose the XGBoost in this case. Bayesian optimization was employed to tune the hyperparameters of the XGBoost model. We got the best Test RMSE: 2.437, with R^2 : 0.419.

Table 3 Bayesian Optimization Results for XGBoost

eta	max_depth	subsample	colsample_bytree	gamma	Value	RMSE	R ²
0.029	4.000	0.959	0.868	0.186	-2.437	2.437	0.419
0.036	3.000	0.719	0.874	0.059	-2.440	2.440	0.418
0.032	3.000	0.888	0.932	0.060	-2.445	2.445	0.416
0.013	4.000	0.615	0.954	0.174	-2.450	2.450	0.413
0.040	3.000	0.677	0.970	0.181	-2.451	2.451	0.413
0.027	3.000	0.740	0.810	0.049	-2.458	2.458	0.409
0.044	5.000	0.625	0.847	0.138	-2.469	2.469	0.404
0.045	3.000	0.881	0.999	0.160	-2.471	2.471	0.403
0.053	3.000	0.723	0.809	0.046	-2.476	2.476	0.401